An Ablation Study of Auxiliary Inputs for Lightweight Neural Upscaling in Real-Time Rendering

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Abstract

This manuscript is in preparation. The abstract will be written after the research question, experimental protocol, and results are finalized.

1 Introduction

Real-time rendering must balance image quality against the time available to produce each frame. Rendering at a lower resolution and reconstructing a higher-resolution image can reduce rendering work, but the reconstruction itself consumes time and must deal with aliased or missing detail [1, 2]. Temporal reconstruction reuses samples from previous frames, while neural methods can combine learned image features with information supplied by the renderer. Both approaches must handle motion and avoid introducing visible artifacts.

Rendering buffers offer geometric and temporal cues beyond color. Prior work has examined their use and ablated components: Wen et al. investigate rendering properties including depth and normals in a framework that also optimizes network size, and Mercier et al. compare temporal reconstruction with and without warping [3, 2]. These studies motivate, but do not determine, the effects of selected inputs in a different model and rendering setup. The quality of a reconstruction also cannot be assessed solely from an aggregate score: for example, Mercier et al. report a benefit of depth-informed motion processing on thin objects even when removing that processing improves aggregate scores in their test [2].

We plan a controlled study of auxiliary inputs for a lightweight neural upscaler. Under a shared experimental protocol, we intend to compare depth and normals individually and together, then compare unaligned with motion-aligned history and examine the joint addition of depth and normals to aligned history. We will assess reconstruction quality alongside measured runtime and memory costs, variation across held-out scenes, and spatial failure cases. This is a proposed comparison within a specific setup, not a claim that input or alignment ablations are new. The architecture, data, and measurement boundaries remain to be fixed before experiments.

2 Background and Related Work

2.1 Efficient Neural Reconstruction

Neural reconstruction can exploit rendering information as well as color, but a more capable model must fit within the rendering time budget. Xiao et al. combine low-resolution color and depth with motion-aligned features from previous frames in a temporal neural supersampling system [1]. Mercier et al. develop an efficient temporal architecture and a gaming dataset with rendered depth and motion; their model-size comparisons illustrate that quality and inference time depend on the selected network and workload [2]. Wen et al. take a different approach, using rendering information to guide network construction, architecture search, and pruning across super-resolution backbones [3]. Thus, comparing their results requires considering both the supplied information and changes in computation. A latency reported for one method does not by itself establish the total cost of another pipeline, including buffer preparation and temporal processing.

2.2 Geometric Guidance

Depth and normals describe different aspects of scene geometry and can be used to guide reconstruction. Wen et al. investigate the sequential addition of rendering properties, including depth, normal direction, and object edges derived from depth, in their CGSR framework [3]. Their analysis relates reconstruction errors to geometric regions and examines accuracy and model size; it also changes the network through its optimization framework. This informs our planned single-frame comparisons of color alone, depth, normals, and their combination, but does not by itself establish their effects in our proposed shared architecture.

Geometric signals can also enter different parts of a temporal model. Zheng et al. use normals together with material information to guide detail reconstruction, while depth and motion guide temporal feature refinement [4]. Their comparison of decoupled guidance with

other fusion designs tests how information is incorporated rather than isolating the effect of each buffer. These approaches motivate comparing individual and combined geometric inputs, while reporting changes in model capacity and cost alongside quality. In particular, adding depth and normals together to aligned history cannot identify their separate effects in the temporal setting.

2.3 Temporal History, Alignment, and Failure Cases

Temporal reconstruction uses motion to relate samples across frames. In Karis's temporal antialiasing method, motion vectors reproject history and neighborhood clipping limits incompatible history [5]. Neural supersampling likewise uses motion-based alignment, but must handle disocclusions and changes that surface motion does not capture, such as changes in lighting [1]. Xiao et al. compare a single-frame model with their temporal model; in a separate comparison they add depth while replacing estimated optical flow with renderer motion vectors. The latter changes both inputs and alignment mechanism, so it does not isolate either effect.

Mercier et al. also compare their temporal method with a no-warp variant, and test depth-informed motion-vector dilation [2]. In their 2× medium-model ablation, removing dilation improves aggregate PSNR and LPIPS, whereas their visual comparison shows reduced artifacts on thin objects with dilation. Xiao et al. illustrate ghosting at disocclusions when their feature-reweighting module is removed [1]. These observations motivate examining localized failures as well as aggregate quality; they do not establish that any planned variant will improve those regions. Our proposed comparison of unaligned and motion-aligned history will test alignment within the selected model, rather than the independent effect of motion vectors or the complete temporal methods above.

3 Method

3.1 Study Design and Analysis Topics

We plan to study a color-only single-frame baseline (V0) and three spatial variants that add depth (V1), normals (V2), or both (V3). A planned temporal extension adds unaligned history to color (V4), aligns that history using motion vectors (V5), and adds depth and normals together to aligned history (V6). Table 1 summarizes the proposed inputs. The temporal model and the representation of history remain to be selected.

Individual and combined geometric inputs. V1 versus V0 estimates the effect of depth, and V2 versus V0 estimates the effect of normals, subject to the capacity and processing controls below. V3 versus V1 estimates the incremental effect of normals with depth, while V3 versus V2 estimates the incremental effect of depth with normals. Together, these comparisons can examine overlap between their contributions in the single-frame setting.

Temporal history and alignment. V4 versus V0 examines adding unaligned history, with any differences in temporal processing taken into account. V5 versus V4 tests motion alignment provided that history and temporal processing are otherwise matched. V6 versus V5 tests adding depth and normals together to aligned history; V6 versus V3 compares the combined temporal and single-frame variants. These comparisons do not isolate the separate effects of depth and normals with aligned history, motion vectors without history, or every interaction among the inputs.

Quality, cost, and generalization. We plan to compare reconstruction quality with the selected runtime and memory measurements. Training, validation, and test scenes will be disjoint at the scene level; test frames will not be used for fitting or protocol selection. We will examine whether variant differences persist across held-out scenes and how they vary among them. An unseen scene does not by itself establish a distribution shift.

Table 1: Planned input variants (V0 is the color-only baseline).

ID	Group	Color	Depth	Normals	History
V0	Single frame	yes	–	–	–
V1	Single frame	yes	yes	–	–
V2	Single frame	yes	–	yes	–
V3	Single frame	yes	yes	yes	–
V4	Temporal	yes	–	–	unaligned
V5	Temporal	yes	–	–	motion aligned
V6	Temporal	yes	yes	yes	motion aligned

Spatial benefits and failures. Alongside aggregate quality, we plan to inspect where variants help or hurt, including geometric edges, thin structures, and regions newly revealed by motion. Error maps and selected image regions can expose local effects that may be obscured in whole-image averages.

Hypotheses. The following are expectations to be assessed after experiments; they may be supported or refuted.

H1 Adding depth improves quality over color-only input.

H2 Adding surface normals improves quality over color-only input.

H3 Depth and normals are partly redundant, so using both improves quality by less than the sum of their individual gains.

H4 History warped with motion vectors improves quality over history used without alignment.

H5 Improvements of selected auxiliary-input variants over their matched baselines persist on scene-disjoint test scenes.

Controls and interpretation. We plan to use a shared base architecture while varying inputs. A shared design does not imply identical parameter counts: input channel changes affect the first layer, and temporal variants may require additional processing. Training data, loss, optimizer, and schedule should be held constant where applicable. We plan multiple random seeds, with the number to be determined. We will report parameter counts and computational complexity, and control capacity where feasible, for example by matching parameter budgets. Any remaining differences in capacity or processing will be acknowledged when interpreting quality and cost. The architecture, temporal implementation, and final protocol remain to be fixed.

3.2 Model, Inputs, and Data

The architecture, input preprocessing, scenes, resolution, and upscaling factor are to be selected. The training, validation, and test scene lists and the method for producing aligned or unaligned history will be fixed before any experiments are run.

3.3 Evaluation Protocol

The following evaluation choices are provisional until the image pipeline and protocol are fixed. No measurements have yet been made. We propose PSNR and SSIM for reference-image reconstruction: PSNR summarizes pixel error, while SSIM compares local image structure [6]. Neither establishes perceptual quality or temporal stability on its own.

For spatial analysis, FLIP is a candidate for perceptually motivated reference-image error maps [7]. If evaluation uses low-dynamic-range (LDR) images, LDR-FLIP applies to the chosen display-ready images. If the study evaluates high-dynamic-range (HDR) images, HDR-FLIP is a distinct method that combines maps over exposure-compensated, tone-mapped image pairs [8]. The choice depends on the final pipeline. FLIP maps indicate estimated visible differences; they do not prove human preferences. LPIPS is an optional complementary learned-feature distance [9], rather than a required metric: its additional value for these rendered images should be established before

inclusion, and its score does not establish human preference. Framewise FLIP and LPIPS do not establish temporal stability.

We will select representative crops and sequences illustrating geometric edges, thin structures, and disocclusions, including cases where inputs help or hurt. Such examples are descriptive. Any systematic regional comparison requires independently defined masks, a selection rule fixed before test analysis, and reporting the number and coverage of regions; if reliable masks cannot be defined, we will not treat selected crops as regional statistics. Metric maps near mask boundaries may depend on context outside the region.

Temporal stability requires a separate sequence-level assessment. As a provisional quantitative option, we may examine temporal variation on verified static segments, alongside the reference sequence; Mercier et al. use pixel-wise temporal standard deviation on a static scene segment [2]. This can reveal flicker in that restricted setting but does not characterize moving content, and a stable blurred output may score well. We also plan synchronized reference/output video inspection for moving scenes, including disocclusions, using predeclared selection and playback conditions. A motion-compensated temporal error may be considered if reliable correspondences and visibility masks are available: its motion source, warping and exclusions must be fixed independently of the variants. Warping errors, occlusion, lighting changes, and blur can confound the result; excluded pixels and coverage must be reported. Neither this measure nor selected videos alone would establish general temporal perceptual quality.

Before testing, the protocol must specify implementation and version for each measure; reference alignment, frame synchronization, resolution and cropping; color space, transfer function, range, exposure and tone mapping; and the treatment of invalid pixels. For PSNR, specify peak value, channels, and whether aggregation precedes or follows the logarithm; for SSIM, specify window, scale, channels and range. For FLIP, specify the LDR or HDR variant, display and viewing assumptions, map pooling, and, for HDR, exposure and tone-mapping settings. If LPIPS is used, specify its backbone, weights, input normalization and pooling. For temporal assessment, specify static-segment criteria, frame rate, duration, reset handling and any correspondence and masking procedure. Fix aggregation across pixels, frames, sequences, scenes, and seeds; report per-scene results and the weighting used for summaries. These choices must apply consistently to the variants.

Possible deployment costs include neural inference latency and GPU memory consumption, as well as buffer preparation, history warping and storage, data transfers, and total upscaling-pipeline time. The protocol must define measurement boundaries, hardware, timing procedure and unmeasured costs. Evaluation metric computation, reference processing, and offline inspection are separate from deployment latency and must not be included in it unless explicitly identified as a separate measurement. The measurement protocol and any interpolation baseline remain to be fixed before the experiments.

4 Results

Results have not yet been collected.

4.1 Individual and Combined Geometric Inputs

4.2 Temporal History and Alignment

4.3 Quality, Runtime, and Memory Trade-offs

4.4 Generalization to Held-out Scenes

4.5 Spatial Benefits and Failure Cases

5 Discussion

This section will interpret the analysis topics and their trade-offs, state whether hypotheses H1 to H5 were supported by the results, and discuss the limitations of the study.

6 Conclusion

Conclusions will be written after the analysis.

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